

Maxima / minima

First Principle of Derivatives:

Theorem : (First Derivative test)

Suppose f is continuous at a critical number x_0 .

(i) If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0 , then f has a relative maximum at x_0 .

(ii) If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending right from x_0 , then f has a relative minimum at x_0 .

(iii) If $f'(x)$ has the same sign [either $f'(x) > 0$ or $f'(x) < 0$] on an open interval extending left from x_0 and on an open interval extending right from x_0 , then f does not have a relative extremum (maximum/minimum) at x_0 .

Theorem: (Second derivative test)

Suppose that f is twice differentiable at x_0 .

- (i) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at x_0 .
- (ii) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at x_0 .
- (iii) If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test is inconclusive (not leading a definite decision), that is, f may have a relative maximum, a relative minimum or neither.

Ex: Find the relative extrema (maxima/minima) using both the first and second derivative tests for $f(x) = x^4 - 8x^2 + 16$

Soln Given, $f(x) = x^4 - 8x^2 + 16$

$$\therefore f'(x) = 4x^3 - 16x = (x)^2$$

$$= 4x(x^2 - 4) = (x)^2$$

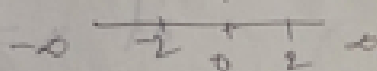
$$= 4x(x-2)(x+2)$$

For critical point,

$$f'(x) = 0$$

$$\Rightarrow 4x(x-2)(x+2) = 0$$

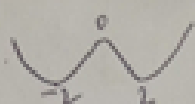
$$\Rightarrow x = 0, -2, 2$$



Intervals: $(-\infty, -2)$, $(-2, 0)$, $(0, 2)$, $(2, +\infty)$

1st derivative test:

Intervals	$4x(x-2)(x+2)$	Sign of $f'(x)$	Conclusion
$(-\infty, -2)$	$(+)(-)(-)(-)$	-	decreasing
$(-2, 0)$	$(+)(-)(-)(+)$	+	increasing
$(0, 2)$	$(+)(+)(-)(+)$	-	decreasing
$(2, +\infty)$	$(+)(+)(+)(+)$	+	increasing



at $x = -2$, $f(x)$ has a relative minima

at $x = 0$, $f(x)$ has a relative maxima

at $x = 2$, $f(x)$ has a relative minima

2nd derivative test:

$$f(x) = x^4 - 8x^2 + 16$$

$$\therefore f'(x) = 4x^3 - 16x = 4x(x-2)(x+2)$$

$$f''(x) = 12x^2 - 16$$

Now, for critical point $f'(x) = 0$

$$\Rightarrow x = 0, -2, 2$$

$$\text{at } x = 0, f''(x) = 0 - 16 < 0$$

$$\text{at } x = -2, f''(x) = 48 - 16 > 0$$

$$\text{at } x = 2, f''(x) = 48 - 16 > 0$$

Therefore, at $x = 0$, there is a relative maxima

at $x = 2, -2$, $f(x)$ has a relative minima.